

# Statistics Practice Set — S19

S19 will focus on **interpreting statistical results and making business decisions** rather than just calculating values.

No Population/Sample Q1 and no formula-recall questions.

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## Q1 — Mean, Median & Business Decision

A company records customer order values:

| Order Value |
|-------------|
| ₹450        |
| ₹480        |
| ₹500        |
| ₹520        |
| ₹550        |
| ₹600        |
| ₹4,500      |

**Answer:**

1. Calculate the mean.
  2. Calculate the median.
  3. Which measure better represents the typical order value?
  4. Explain why.
  5. What effect does ₹4,500 have on the mean?
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## Q2 — SD vs CV

Two factories have:

| Factory | Mean Production | SD  |
|---------|-----------------|-----|
| A       | 1,000 units     | 80  |
| B       | 4,000 units     | 200 |

**Answer:**

1. Calculate CV for both.
  2. Which factory has greater absolute variability?
  3. Which factory has greater relative variability?
  4. Which factory is more consistent relative to its production level?
  5. Why would comparing SD alone give an incomplete picture?
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## Q3 — IQR vs Range

Two datasets represent delivery times:

**Dataset A:**

20, 22, 24, 25, 26, 28, 30

**Dataset B:**

20, 22, 24, 25, 26, 28, 100

**Answer:**

1. Calculate the range of both datasets.
  2. Calculate Q1 and Q3 for both.
  3. Calculate the IQR of both.
  4. Which measure, Range or IQR, is better for comparing the typical spread?
  5. Explain why the two measures behave differently.
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**Q4 — Skewness From Summary Statistics**

A dataset has:

| Statistic | Value |
|-----------|-------|
| Minimum   | 10    |
| Q1        | 18    |
| Median    | 25    |
| Q3        | 30    |
| Maximum   | 75    |

**Answer:**

- 1. Calculate the lower whisker length.
  - 2. Calculate the upper whisker length.
  - 3. Which side has the longer tail?
  - 4. Identify the likely skewness.
  - 5. Give **two pieces of evidence** for your answer.
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**Q5 — Box Plot Business Comparison**

Two sales teams have:

| Statistic | Team A | Team B |
|-----------|--------|--------|
| Minimum   | ₹20k   | ₹10k   |
| Q1        | ₹30k   | ₹20k   |
| Median    | ₹40k   | ₹35k   |
| Q3        | ₹50k   | ₹60k   |
| Maximum   | ₹60k   | ₹100k  |

**Answer:**

1. Which team has the higher median?
  2. Calculate IQR for both.
  3. Calculate range for both.
  4. Which team has greater variability?
  5. Which team appears more consistent?
  6. Which team has the longer upper tail?
  7. If the company wants **stable sales performance**, which team would you prefer?
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**Q6 — Outlier Detection & Business Impact**

A delivery dataset contains:

**15, 16, 18, 19, 20, 21, 22, 24, 26, 70**

**Using the IQR method:**

1. Find Q1.
  2. Find Q3.
  3. Find IQR.
  4. Find the lower fence.
  5. Find the upper fence.
  6. Identify the outlier.
  7. Would you use the mean or median to describe the typical delivery time?
  8. Explain why.
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### Q7 — CV Ranking & Decision

Three investment options have:

| Investment | Mean Return | SD   |
|------------|-------------|------|
| A          | 10%         | 2%   |
| B          | 18%         | 3.6% |
| C          | 30%         | 4.5% |

**Answer:**

1. Calculate CV for each.
  2. Rank them from **most consistent** → **least consistent**.
  3. Which has the highest return?
  4. Which has the lowest relative variability?
  5. Does the highest-return investment also have the lowest CV?
  6. If an investor prioritizes **relative consistency**, which option is best?
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### Q8 — Identify the Weak Analyst Conclusion

An analyst says:

**“Dataset A has a higher mean than Dataset B, so Dataset A must have higher customer spending for most customers.”**

Is this conclusion valid?

**Answer:**

1. Yes or No?
  2. Explain the problem with this reasoning.
  3. Which additional statistic would you check?
  4. Give an example where the mean could be higher but the typical customer actually spends less.
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### Q9 — Choose the Best Statistical Tool

Choose the **best option** for each scenario.

#### Options:

- Mean
- Median
- SD
- IQR
- CV
- Box plot

#### A.

A company wants the typical salary when a few executives earn extremely high salaries.

#### B.

A manager wants to compare relative consistency between two stores with very different average sales.

#### C.

An analyst wants to measure the spread of the middle 50%.

#### D.

An analyst wants to visually compare six departments and identify possible outliers.

#### E.

Two teams have the same average delivery time, and the manager wants to know which has more absolute variation.

#### F.

A dataset has no major extreme values and the analyst wants the arithmetic average.

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### Q10 — Full Data Analyst Case

Two e-commerce platforms have the following order statistics:

| Platform | Mean Order Value | Median | SD   |
|----------|------------------|--------|------|
| A        | ₹1,200           | ₹900   | ₹240 |
| B        | ₹2,000           | ₹1,950 | ₹300 |

#### Answer:

1. Calculate CV for both.
2. Which platform has higher absolute variability?
3. Which platform has higher relative variability?
4. Which platform appears more right-skewed?
5. Which platform has more evidence that extreme high orders are affecting the mean?
6. If the company wants **higher consistency relative to average order value**, which platform should it choose?
7. If the company wants **lower absolute variation in order values**, which platform should it choose?
8. Explain why looking at **mean + median + SD + CV together** gives a better picture than using only one statistic.